

Autonomous Forest Drone

Developer Handover Document

Elina Rosato, Danielis Maizelis

Kristianstad University, Sweden

April 2026

Contents

1	Introduction	4
2	System Architecture	4
3	Hardware Assembly	4
3.1	Bill of Materials	4
3.1.1	Purchased Components	4
3.1.2	3D Printed Parts	5
3.2	Tools Required	6
3.3	Assembly Guide	6
3.3.1	Step 1 – Frame	6
3.3.2	Step 2 – Motors	6
3.3.3	Step 3 – Propeller Guards	7
3.3.4	Step 4 – Battery Mount and Power Module	7
3.3.5	Step 5 – Jetson Orin Nano	7
3.3.6	Step 6 – Pixhawk with Vibration Dampening	7
3.3.7	Step 7 – Telemetry Radio	7
3.3.8	Step 8 – RC Receiver	8
3.3.9	Step 9 – Camera Mount and Camera	8
3.3.10	Step 10 – RC Receiver Binding	8
3.3.11	Step 11 – GPS Mast and GPS Module	8
4	Wiring Guide	9
5	Pixhawk Configuration	10
5.1	Firmware and Airframe	10
5.2	Battery Configuration	10
5.3	Calibration Sequence	10
5.4	Failsafe Configuration	11

5.5	Parameter Configuration	11
6	Software Setup	11
6.1	Jetson Orin Nano Environment	11
6.2	ROS 2 Humble and MAVROS	11
6.3	Camera Setup (Arducam IMX219)	12
6.4	drone-brain Repository	12
6.5	ground-control-station Repository	13
7	Commissioning and Testing	13
7.1	Safety Setup for Outdoor Testing	13
7.2	Test Sequence	14
7.2.1	Test 1 – Motor Test (No Propellers)	14
7.2.2	Test 2 – RC Channel Verification	14
7.2.3	Test 3 – Arm-Only Script	14
7.2.4	Test 4 – Takeoff with Manual Intercept	14
7.2.5	Test 5 – Takeoff and Land	15
7.2.6	Test 6 – Takeoff, Move Forward, Land	15
7.2.7	Test 7 – Full Movement Test	15
7.2.8	Test 8 – GPS Position Navigation	15
7.2.9	Test 9 – Camera Tests	15
8	Known Limitations	16
8.1	GPS Accuracy Under Forest Canopy	16
8.2	Weather Dependency	16
9	Future Work	16
9.1	Positioning Under Canopy	16
9.2	Image Transmission	16
10	Repository and Access Summary	17

1 Introduction

This document provides the information necessary for a developer or researcher to take over the Autonomous Forest Drone project and continue its development. It covers the full system from hardware assembly to software setup and progressive commissioning, and should be read alongside the companion **User Manual**, which covers day-to-day operation, pre-flight procedures, and troubleshooting.

Scope of this document:

- Physical assembly of the drone from parts
- All wiring connections with connector types on both ends
- Pixhawk firmware flashing, calibration, and parameter configuration
- Software environment setup on the Jetson
- Repository setup and architecture
- Progressive commissioning and testing sequence
- Known limitations and suggested future work

GitHub organisation: <https://github.com/autonomous-forest-drone>

2 System Architecture

The system consists of three layers:

1. **Flight platform:** Holybro S500 V2 quadcopter frame with Pixhawk 6C running PX4 firmware. The Pixhawk handles all low-level flight control, sensor fusion, and safety features.
2. **Companion computer:** NVIDIA Jetson Orin Nano 8 GB running ROS 2 Humble with MAVROS. The Jetson executes autonomous mission scripts that communicate with the Pixhawk over MAVLink via a UART serial link (Jetson GPIO TX/RX pins to Pixhawk TELEM2).
3. **Ground station:** A laptop running QGroundControl for monitoring and parameter management, and the `ground-control-station` repository scripts for receiving telemetry from the drone.

3 Hardware Assembly

3.1 Bill of Materials

All components required to build and operate the drone are listed below.

3.1.1 Purchased Components

Component	Part Name / Model	Notes
Frame kit	Holybro S500 V2 Quadcopter Kit	Includes frame, arms, landing gear, PDB, motors, ESCs, and hardware
Flight controller	Holybro Pixhawk 6C	
GPS module	Holybro M10 GPS with compass	
Power module	Holybro PM02	Voltage and current sensing
Telemetry radio (air)	RFD900x or SiK Radio	915 MHz or 433 MHz
Telemetry radio (ground)	Matching RFD900x or SiK unit	Must match frequency band
RC receiver	FlySky FS-iA6B	PPM mode
RC transmitter	FlySky FS-i6	6-channel, 2.4 GHz AFHDS
Companion computer	NVIDIA Jetson Orin Nano 8 GB Developer Kit	Includes case
Camera	Arducam IMX219	8 MP, 62° FOV, MIPI CSI, fixed focus
Main battery	4S LiPo 14.8 V 5000 mAh	XT60 connector
Battery charger	4S LiPo balance charger	Balance charging required
XT60 Y-splitter cable	XT60 female to 2× XT60 male	Splits battery to power module and Jetson supply
Jetson power adapter cable	XT60 male to 5.5×2.5 mm barrel jack	Direct connection from Y-splitter to Jetson Orin Nano power input; no converter needed (Jetson accepts 9–20 V)
Vibration dampening plate	M3-compatible foam or rubber standoff plate	Between Pixhawk and Jetson case
CSI ribbon cable	22-pin (Jetson) to 15-pin (camera), 0.5 mm pitch	Arducam IMX219 to Jetson CSI0 (camera0)
Jetson–Pixhawk UART cable	Female Dupont 3-pin (Jetson GPIO) to JST-GH 6-pin (Pixhawk TELEM2)	Custom cable; TX, RX, GND
Motor mounting screws (extended)	M3 × (stock length + 6 mm)	See Section 3.3.3; do not exceed this length
Velcro straps (×2)	Heavy-duty adhesive velcro	Battery and Jetson mounting
Zip ties	Assorted	Cable management

3.1.2 3D Printed Parts

The following parts were designed and 3D printed for this build. Print files are stored in the project GitHub organisation.

Part	Quantity	Notes
Propeller guard	4	One per motor arm
Camera mount	1	Attaches to front of frame; holds Arducam IMX219 facing forward
GPS mast	1	Custom height required – the original Holybro mast is too short when the Jetson is installed. The mast must raise the GPS module high enough to clear the Jetson and its case to avoid magnetic interference.

3.2 Tools Required

- Hex key set (M2, M2.5, M3)
- Philips and flathead screwdrivers
- Soldering iron and solder (for any power connector modifications)
- Digital multimeter (continuity and voltage checks)
- LiPo voltage checker (balance connector)
- 3D printer (for the custom parts listed above)
- Computer with QGroundControl installed

3.3 Assembly Guide

Assemble the drone in the order described below. Do not connect the main battery until Section 7.2.1.

3.3.1 Step 1 – Frame

Assemble the Holybro S500 V2 frame following the kit instructions:

1. Attach the four arms to the central frame body.
2. Attach the landing gear legs to the underside.
3. Mount the Power Distribution Board (PDB) in the centre of the frame. The PDB is included in the S500 kit.

3.3.2 Step 2 – Motors

1. Install the four Holybro 2216 880KV brushless motors on the arm motor mounts following the CW/CCW positions specified for the S500 V2 kit (see the motor diagram in the User Manual, Appendix D).
2. **Do not fully tighten the motor screws yet** – the propeller guards are installed in the next step and require replacing the screws.

3.3.3 Step 3 – Propeller Guards

1. 3D print four propeller guards.
2. Place one guard around each motor mount before the final motor screw installation.
3. The guards add thickness between the motor and the arm. The standard M3 motor screws included in the S500 kit are too short to grip the motor securely with the guard in place. **Replace the motor screws with M3 screws that are 6 mm longer than the stock screws.**
4. **Important:** Do not use screws longer than necessary. If the screw protrudes too far into the motor casing, it will damage the motor windings. 6 mm of additional length is the correct amount for the printed guards used in this build.
5. Tighten all motor screws with a hex key.

3.3.4 Step 4 – Battery Mount and Power Module

1. Attach adhesive velcro to the battery holder on the underside of the frame and to the battery.
2. Mount the Holybro PM02 Power Module between the battery output and the PDB. The power module has an XT60 female input (battery side) and an XT60 male output (PDB side). It also has a 6-pin JST-GH cable that connects to the Pixhawk POWER1 port for voltage and current sensing.
3. The battery connects to the power module via the XT60 Y-splitter (see Section 4).

3.3.5 Step 5 – Jetson Orin Nano

1. Attach heavy-duty adhesive velcro to the top of the central frame body and to the bottom of the Jetson Orin Nano case.
2. Seat the Jetson (in its case) on the velcro on top of the frame. Centre it for weight balance.
3. Route the Jetson power adapter cable (XT60 to barrel jack) to the rear of the frame where the XT60 Y-splitter will be accessible. The Jetson Orin Nano Developer Kit accepts 9–20 V directly, so the LiPo voltage (14.8 V nominal) is within spec with no converter needed.

3.3.6 Step 6 – Pixhawk with Vibration Dampening

1. Attach the vibration dampening plate to the top of the Jetson case using the adhesive on the plate.
2. Mount the Pixhawk 6C on top of the vibration dampening plate. The arrow on the Pixhawk must point toward the front of the drone.
3. Secure the Pixhawk to the plate using double sided tape.

3.3.7 Step 7 – Telemetry Radio

Mount the air-side telemetry radio (RFD900x or SiK) on the frame using velcro or a tie. Position the antenna so it points vertically downward when the drone is in flight for optimal ground link. Route the cable to the Pixhawk TELEM1 port.

3.3.8 Step 8 – RC Receiver

Mount the FlySky FS-iA6B RC receiver on the frame with its antenna wires routed away from the ESC power cables to minimise RF interference. Route the PPM signal cable to the Pixhawk RC IN port.

3.3.9 Step 9 – Camera Mount and Camera

1. 3D print the camera mount.
2. Install the Arducam IMX219 camera in the mount with the lens facing forward.
3. Attach the mount to the front of the frame.
4. Route the 22-pin to 15-pin CSI ribbon cable from the camera toward the Jetson CSI0 connector (camera0). Take care not to kink or crease the ribbon cable; leave a gentle loop for vibration relief.

3.3.10 Step 10 – RC Receiver Binding

Bind the FlySky FS-iA6B receiver to the FlySky FS-i6 transmitter before permanently routing cables:

1. Power the receiver using a temporary 5 V source (e.g. connect it to the Pixhawk RC IN port and power the Pixhawk via USB from a computer).
2. Hold the bind button on the receiver while powering it on. The LED will flash rapidly, indicating bind mode.
3. On the FS-i6 transmitter, navigate to **System** → **RX Bind** and confirm. The transmitter emits a bind signal.
4. Once bound, the receiver LED turns solid. Power cycle the receiver to confirm the bind is stored.
5. Confirm the receiver is set to **PPM mode**: on power-on, the LED should flash twice. One flash indicates PWM mode – switch to PPM by pressing the bind button briefly after power-on (consult the FS-iA6B manual for the exact procedure for your firmware version).

3.3.11 Step 11 – GPS Mast and GPS Module

1. 3D print the custom GPS mast. The mast must be tall enough to position the GPS module above and clear of the Jetson Orin Nano. The original Holybro mast was found to be too short; the Jetson creates magnetic interference that corrupts the compass if the GPS is too close to it.
2. Attach the mast to the rear-top of the frame.
3. Mount the Holybro M10 GPS module on top of the mast with the arrow pointing forward (same as the Pixhawk orientation).
4. Route the GPS cable down the mast and to the Pixhawk GPS1 port.

4 Wiring Guide

Complete all wiring after the physical assembly is done. Refer to the connection diagram in the User Manual for a visual overview. The table below lists every cable with the connector type on both ends.

Cable	From	To	Connector
Battery to Y-splitter	4S LiPo (female XT60)	XT60 Y-splitter input	XT60
Y-splitter to power module	XT60 Y-splitter output 1	PM02 Power Module input	XT60
Y-splitter to Jetson	XT60 Y-splitter output 2	Jetson power adapter (XT60 male end)	XT60
Jetson power	Jetson power adapter (barrel end)	Jetson Orin Nano power jack	5.5×2.5 mm barrel
Power module to Pixhawk	PM02 telemetry output	Pixhawk 6C POWER1	6-pin JST-GH
Power module to PDB	PM02 output	PDB XT60 input	XT60
ESC 1–4 signal wires	Each ESC signal lead (female Dupont/servo)	PWM signal distribution board channels 1–4	Servo 3-pin (signal/+ /GND)
PWM board to Pixhawk	PWM distribution board output	Pixhawk 6C MAIN OUT 1–4	Servo rail / JST
GPS	Holybro M10 GPS	Pixhawk 6C GPS1	8-pin JST-GH
Telemetry radio (air)	RFD900x / SiK air unit	Pixhawk 6C TELEM1	6-pin JST-GH
RC receiver	FlySky FS-iA6B PPM output	Pixhawk 6C RC IN	3-pin servo (PPM)
Jetson to Pixhawk (MAVLink)	Jetson GPIO TX/RX/GND (Dupont female)	Pixhawk 6C TELEM2 (JST-GH 6-pin)	Custom UART cable
Camera CSI	Arducam IMX219	Jetson Orin Nano CSI0 (camera0)	22-pin to 15-pin ribbon

Notes on the ESC wiring:

The S500 kit ESCs each have a three-wire servo signal lead (signal, +5 V BEC, GND). These were connected to a PWM signal distribution board (a small PCB with male pin headers for each channel). Channels 1–4 of that board receive the four ESC signal leads, and the board’s output connector plugs into the Pixhawk MAIN OUT rail. This avoids routing four individual servo wires directly to the Pixhawk.

Motor-to-ESC power wiring (the thick 3-phase cables) is part of the S500 kit and soldered from the factory; verify continuity if any cable has been disturbed.

Notes on the Jetson–Pixhawk UART link:

The Jetson connects to the Pixhawk via a custom UART cable: Jetson GPIO TX and RX pins (40-pin header) to Pixhawk TELEM2 (JST-GH 6-pin). Wire TX on the Jetson to RX on the Pixhawk, RX on the Jetson to TX on the Pixhawk, and GND to GND. On the Jetson Orin Nano, the UART appears as `/dev/ttyTHS1`. The PX4 TELEM2 baud rate must be set to 115200 via the `SER.TEL2.BAUD` parameter (see Section 5.5). If the MAVROS connection fails on startup, verify the device node exists with `ls /dev/ttyTHS*` and that the user has serial port permissions (`sudo usermod -aG dialout $USER`, then log out and back in).

Cable management:

After verifying all connections, use zip ties to bundle cables neatly along the frame arms and body. Keep high-current power cables (battery, ESC, motor) separated from signal cables (GPS, RC, telemetry, CSI) to minimise interference.

5 Pixhawk Configuration

5.1 Firmware and Airframe

1. Download and install QGroundControl from <https://qgroundcontrol.com>.
2. Connect the Pixhawk to a computer via the USB-C cable. **Do not connect the main battery.**
3. In QGC, go to **Vehicle Setup** → **Firmware**. Select **PX4 Flight Stack – Stable Release** and confirm the flash. Wait for the process to complete and the Pixhawk to reboot.
4. After flashing, QGC prompts for airframe selection. Choose **Holybro S500** (under the Quadro-rotor X category). Click **Apply and Restart**.

Important: Flashing firmware resets all parameters to defaults. Always run the full calibration sequence and re-apply the custom parameters after any reflash.

5.2 Battery Configuration

In QGC, go to **Vehicle Setup** → **Power** and set the battery fields to match the 4S 5000mAh LiPo:

Field	Value
Number of cells	4
Full voltage (per cell)	4.20 V
Empty voltage (per cell)	3.50 V

5.3 Calibration Sequence

Run the calibrations in the order listed below via **Vehicle Setup** → **Sensors**.

1. **Level Horizon** – Place the drone on a level surface and click **Level Horizon**.
2. **Accelerometer (6-Position)** – Follow the on-screen prompts. Place the drone in each of the six orientations indicated (level, nose down, nose up, right side down, left side down, upside down) and confirm each position.
3. **Gyroscope** – Place the drone on a still, level surface. Do not move it during calibration.
4. **Compass** – Perform outdoors, at least 2 m from metal objects, buildings, and electronics. Follow the on-screen sphere visualisation, rotating the drone through all axes until all coloured segments are filled.
5. **ESC Calibration** – Remove all propellers. Go to **Vehicle Setup** → **Power** → **ESC Calibration**. Follow the QGC wizard: set the RC throttle to maximum, connect the main battery, wait for confirmation beeps from the ESCs, then lower throttle to minimum for a second set of beeps. Disconnect the battery when complete.

6. **RC Radio** – Go to **Vehicle Setup** → **Radio**. Move all sticks to their full extents and all switches through all positions. Click **Next** to save.
7. **Flight Modes** – Go to **Vehicle Setup** → **Flight Modes**. Assign Channel 5 (3-position switch) to cycle through Stabilise, Altitude Hold, and Position modes. This switch is also used to cancel Offboard mode during autonomous flight.

5.4 Failsafe Configuration

In QGC, go to **Vehicle Setup** → **Safety** and configure:

Failsafe	Setting
RC loss action	Return to Launch (RTL) after 1 s
Low battery action	Return to Launch at 20%; Land at 10%

5.5 Parameter Configuration

After calibration, apply the project-specific parameters. A parameter file (`.params` export) is stored in the `drone-brain` repository (<https://github.com/autonomous-forest-drone>).

To apply parameters from the file:

1. In QGC, go to **Vehicle Setup** → **Parameters**.
2. Click the **Tools** menu (top right) and select **Load from file**.
3. Select the `.params` file from the repository.
4. Click **Save** and reboot the Pixhawk when prompted.

6 Software Setup

6.1 Jetson Orin Nano Environment

The Jetson Orin Nano Developer Kit ships with JetPack (Ubuntu 22.04 + CUDA). Before setting up the project software, verify the OS is up to date:

```
1 sudo apt update && sudo apt upgrade -y
```

6.2 ROS 2 Humble and MAVROS

ROS 2 Humble must be installed on the Jetson and is the communication backbone between the mission scripts and the Pixhawk.

Install **ROS 2 Humble** following the official instructions at <https://docs.ros.org/en/humble/Installation/Ubuntu-Install-Debian.html>.

After installing ROS 2, install MAVROS and its dependencies:

```
1 sudo apt install ros-humble-mavros ros-humble-mavros-extras
2 ros2 run mavros install_geographiclib_datasets.sh
```

Add the ROS 2 setup to the Jetson's shell profile so it is sourced automatically:

```
1 echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
2 source ~/.bashrc
```

Grant UART port permissions for the Jetson–Pixhawk link:

```
1 # Verify the UART device is present
2 ls /dev/ttyTHS*
3 # Add user to dialout group for persistent access
4 sudo usermod -aG dialout $USER
5 # Log out and back in, then verify:
6 ls -l /dev/ttyTHS1
```

6.3 Camera Setup (Arducam IMX219)

The Arducam IMX219 connects to the Jetson CSI0 port via the 22-pin to 15-pin ribbon cable. Install the Arducam driver [5]:

```
1 # Follow the official Arducam Jetson installation guide for the IMX219
2 # https://docs.arducam.com/Nvidia-Jetson-Camera/Native-Camera/Quick-Start-Guide/
```

Verify the camera is working with GStreamer:

```
1 gst-launch-1.0 nvarguscamerasrc ! nvoverlaysink
```

6.4 drone-brain Repository

The `drone-brain` repository contains all autonomous mission scripts, pre-flight tests, and helper utilities. It runs on the Jetson Orin Nano.

Clone and install:

```
1 git clone https://github.com/autonomous-forest-drone/drone-brain.git ~/drone-brain
2 cd ~/drone-brain
3 pip3 install -r requirements.txt
```

Repository structure:

Directory / File	Contents
<code>missions/</code>	Autonomous flight mission scripts. Each script encodes a complete flight plan and is intended to be run in the field.
<code>tests/</code>	Commissioning and pre-flight test scripts. See Section 7 for the required test sequence.
<code>helpers/</code>	Utility scripts that run alongside a mission in a separate terminal. Includes telemetry monitoring and image transmission.
<code>docs/</code>	Additional developer documentation specific to the repository.
<code>requirements.txt</code>	Python dependencies.
<code>README.md</code>	Entry point for the repository. Always read this first – it is the authoritative source for any setup steps not covered here.

All scripts follow the same two-terminal pattern:

Terminal 1 – start MAVROS (required for every run):

```
1 ros2 launch mavros px4.launch fcu_url:=/dev/ttyTHS1:115200
```

Terminal 2 – run a script:

```
1 python3 ~/drone-brain/missions/<mission>.py
2 python3 ~/drone-brain/tests/<test>.py
3 python3 ~/drone-brain/helpers/<helper>.py
```

Use `tmux` to keep sessions alive over SSH:

```
1 tmux new -s drone
2 # Split: Ctrl+B then %
3 # Detach: Ctrl+B then D
4 # Reattach: tmux attach -t drone
```

6.5 ground-control-station Repository

The `ground-control-station` repository runs on the monitoring laptop. It receives and displays data transmitted by the drone.

```
1 git clone https://github.com/autonomous-forest-drone/ground-control-station.git \
2   ~/ground-control-station
3 cd ~/ground-control-station
4 pip3 install -r requirements.txt
```

Refer to the repository README for the current list of available scripts. At the time of writing, it includes an `image-receiver` script that displays images transmitted by the drone's `image-sender` helper.

7 Commissioning and Testing

After completing hardware assembly, wiring, Pixhawk configuration, and software setup, commission the drone by working through the test sequence below in order. Do not advance to the next test until the current one passes cleanly.

7.1 Safety Setup for Outdoor Testing

For all initial outdoor tests, use physical restraints to limit flyaway risk:

- Drive four pegs into the ground at the corners of a roughly 1 m square centred on the drone's takeoff position.
- Attach a rope or strong cord from each landing leg to a peg, leaving enough slack for the drone to lift approximately 0.5–1 m off the ground.
- Ensure all ropes are free of the propeller arc.
- Keep one operator on the RC transmitter at all times, ready to switch to Altitude/Position mode.

7.2 Test Sequence

7.2.1 Test 1 – Motor Test (No Propellers)

With propellers removed and the drone connected to QGC via USB:

1. Go to **Vehicle Setup** → **Actuators** → **Motors**.
2. Spin each motor individually at low throttle using the sliders.
3. Verify each motor spins in the correct direction (see motor diagram in the User Manual, Appendix D): M1 Front-Right CCW, M2 Back-Left CW, M3 Front-Left CW, M4 Back-Right CCW.
4. If a motor spins in the wrong direction, swap any two of its three phase wires at the ESC output.

7.2.2 Test 2 – RC Channel Verification

With the RC transmitter powered on and QGC connected:

1. Go to **Vehicle Setup** → **Radio**.
2. Move each stick and switch on the FS-i6 and confirm the corresponding channel bar moves in QGC.
3. Verify stick directions match expected directions (e.g., throttle up = Channel 3 increases).
4. Confirm Channel 5 (3-position switch) cycles through the three flight modes and Channel 6 (arming switch) responds.

7.2.3 Test 3 – Arm-Only Script

Install propellers. Move outdoors. Use the rope and peg safety setup.

1. SSH into the Jetson and start MAVROS (Terminal 1).
2. In Terminal 2, run the arm-only test script from the `tests/` directory.
3. The drone should arm (motors spin at idle) and then disarm after a short delay without taking off.
4. Verify QGC shows ARMED then DISARMED and no errors appear.

7.2.4 Test 4 – Takeoff with Manual Intercept

1. Run the takeoff-only script from `tests/`.
2. The drone lifts to a set hover altitude. **At any point**, switch the RC transmitter to Stabilise mode to cancel Offboard and take manual control.
3. Practice the manual intercept several times to confirm the safety override works reliably before proceeding.

7.2.5 Test 5 – Takeoff and Land

1. Run the takeoff-and-land script. The drone should arm, climb to hover altitude, hold position for a few seconds, descend, land, and disarm automatically.
2. Verify there is no significant horizontal drift during hover.

7.2.6 Test 6 – Takeoff, Move Forward, Land

1. Run the forward-movement script. The drone should take off, translate a short distance forward, then return and land.
2. Verify the movement is smooth and the drone returns to approximately the launch position.
3. This test validates that the Offboard velocity commands are working correctly.

7.2.7 Test 7 – Full Movement Test

1. Run the movement-test script. This test takes off and performs controlled movements on all axes in sequence: forward, backward, left, right, yaw left, yaw right, and altitude step, then lands.
2. Verify all movements are smooth and the drone responds correctly on each axis.
3. This is the primary test to validate full Offboard control before GPS-based navigation.

7.2.8 Test 8 – GPS Position Navigation

1. Physically carry the drone to the intended goal position. Run the goal-setting command with the `--set-goal` flag to record the GPS coordinates:

```
1 python3 ~/drone-brain/tests/<gps-nav-test>.py --set-goal
```

2. Return the drone to the launch position.
3. Run the GPS navigation test (without `--set-goal`). The drone should navigate autonomously to the recorded coordinates, hold for a moment, and return to launch.
4. Refer to the test script file and its inline documentation for the exact flags and behaviour.

7.2.9 Test 9 – Camera Tests

1. Run the camera test scripts from `tests/` to verify image capture. These can be run on the ground with the drone stationary.
2. Check that captured frames are correctly saved or transmitted.
3. Run the telemetry monitoring helper and confirm live telemetry messages are printing to the terminal and/or appearing on the ground station.

8 Known Limitations

8.1 GPS Accuracy Under Forest Canopy

The Holybro M10 GPS has a position accuracy of approximately 2.5 m under good open-sky conditions. Under forest canopy, satellite reception degrades significantly:

- The GPS position becomes **noisy and unreliable** under dense tree cover. Position errors can exceed several metres and fluctuate rapidly.
- The PX4 EKF fuses GPS with the barometer and IMU, but sustained GPS noise causes position hold drift in Offboard mode.
- Waypoint-based navigation in forest conditions therefore produces imprecise results and is not suitable for close-range obstacle avoidance.

This is a fundamental hardware limitation of single-frequency GNSS in obstructed environments, not a software issue. Addressing it would require a different positioning approach (see Section 9).

8.2 Weather Dependency

The drone has no weatherproofing. Rain, fog, or high humidity can damage electronics and corrupt sensor readings. Maximum sustained wind for safe autonomous operation is 3 m/s.

9 Future Work

The following directions are the most relevant for a developer continuing this project.

9.1 Positioning Under Canopy

Replacing or augmenting the GPS with sensors that work under tree cover is the most impactful improvement available:

- **Optical flow sensor** (e.g. PX4FLOW or equivalent) – measures ground-relative velocity from a downward-facing camera. Fuses with the barometer in PX4 EKF to provide position hold without GPS. Works indoors and under canopy but drifts over time.
- **Depth camera or LiDAR** (e.g. Intel RealSense, RPLiDAR) – enables local obstacle detection and SLAM-based localisation on the Jetson.
- **RTK GPS** – provides centimetre-level accuracy outdoors but does not solve the canopy occlusion problem.

9.2 Image Transmission

The current `image-sender` helper transmits 128×128 pixel images over MAVLink Tunnel messages every 20 seconds. Image quality and update rate are limited by the MAVLink channel bandwidth. Alternatives include:

- Transmitting over a dedicated WiFi link (requires the drone to remain within WiFi range).
- Logging full-resolution images to the Jetson’s microSD and retrieving them post-flight.

10 Repository and Access Summary

Resource	Location
GitHub organisation	https://github.com/autonomous-forest-drone
drone-brain (Jetson scripts)	https://github.com/autonomous-forest-drone/drone-brain
ground-control-station	https://github.com/autonomous-forest-drone/ground-control-station
User Manual source	https://github.com/autonomous-forest-drone/user-manual
PX4 parameter file	drone-brain repo root or dedicated config repo in the organisation
3D print files	Project GitHub organisation

References

- [1] PX4 Development Team, *PX4 Autopilot User Guide*, <https://docs.px4.io>, 2024.
- [2] Open Robotics, *ROS 2 Humble Hawksbill Documentation*, <https://docs.ros.org/en/humble/>, 2024.
- [3] MAVROS Contributors, *MAVROS – MAVLink extendable communication node for ROS 2*, <https://github.com/mavlink/mavros>, 2024.
- [4] Holybro, *S500 V2 Development Kit User Guide*, <https://holybro.com/products/s500-v2-development-kit>, 2024.
- [5] Arducam, *IMX219 Camera for NVIDIA Jetson – Quick Start Guide*, <https://docs.arducam.com/Nvidia-Jetson-Camera/Native-Camera/Quick-Start-Guide/>, 2024.